

## Note

### Increasing the Minimum Distance of a Set of Points

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Recently I received a letter from Paul Erdős in which he raised the following problem: Given  $n$  points in  $E^k$  with minimum distance 1, let  $f(n, k)$  be the largest number such that we can be sure to find  $f(n, k)$  points among the  $n$  which have minimum distance  $> 1$ . (Clearly  $f(n, k) \leq [n/(k+1)] + 1$ , by considering  $[n/(k+1)]$  widely spaced unit  $k$ -simplices.) Erdős asks for a proof or disproof of:  $f(n, k) > cn/k$ . Moreover, he reports that he has  $f(n, 2) > n/5$ .

Here is a very short proof that  $f(n, 2) > n/4$ . Note that the distance 1 graph (2 points are adjacent if they have distance 1) of a planar configuration with minimum distance 1 is planar. (Some side of a convex quadrilateral is less than the average of the diagonals.) Then simply four-color this graph, and choose those points which have been colored with the most frequently used color.

*Remark 1.* Janos Pach, and independently Ron Graham and Fan Chung have found a configuration (a chain of 6 double unit  $k$ -simplices) of  $6k + 7$  points from which we can select at most  $k$  points with minimum distance  $> 1$ . Thus  $f(n, k) \leq 6n/(6k + 7) < n/(k + 1)$ .

*Remark 2.* Janos Pach has pointed out that we can do without the four color theorem as a simple induction shows directly that the unit distance graph is four colorable.

#### REFERENCE

1. W. O. J. MOSER AND J. PACH (Eds.), Problem 83, "Research Problems in Discrete Geometry," mimeo, August 1985.

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